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SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

Table 1: Rigor Adherence Table

Ethics
IACUC: All animal procedures used in this study were approved by Boston University Institutional Animal Care and Use Committee (IACUC)
Inclusion and Exclusion Criteria
not detected.
Attrition
not detected.
Sex as a biological variable
Both female and male mice were used , and all mice were 8-12 weeks old at the start of the study .
Subject Demographics
Age: Both female and male mice were used , and all mice were 8-12 weeks old at the start of the study .
Randomization
To determine whether a correlation coefficient value observed between a neuron pair was significantly higher than that expected from random overlap of calcium events in that neuron pair, we used a shuffling procedure to estimate the baseline distribution of correlation values.
Blinding
not detected.
Power Analysis
Through manual inspection we noted that calcium events are associated with increased low frequency (<2Hz) power, therefore, we calculated the power trace as the mean Z scoreZ-score of the power across frequencies below 2Hz.
Replication
This procedure was repeated 1000 times to build the distribution of the shuffled baseline distribution.
Number: This procedure was repeated 1000 times to build the distribution of the shuffled baseline distribution.

Code Information
Availability: Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request EXPERIMENTAL MODEL AND SUBJECT DETAILS Adult GM24Gsat (Chat-cre, MMRRC Stock Number: 017269-UCD), B6 129P2-Pvalbtm1 (cre)ArbrJ (PV-cre, Jax Stock Number: 017320) and C57B6 mice (Charles River Laboratories, Cat C57BL6) were purchased from Charles River Laboratories,
Data Information
Availability: All original code has been deposited at GitHub https:github.comHanLabBUDopamine-Depletion-CellReportsreleasestagv1.0 and Zenodo https:doi.org10.5281zenodo.5786809 and is publicly available as of the date of publication.

Table 2: Key Resources Table

Your Sentences	REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies			
Striatal brain slices were stained using a Rabbit anti-TH antibody (Abcam , No. AB112 , 1:250) followed by either Alexa Fluor 633 goat anti-rabbit secondary antibody (Invitrogen , No. A21070 , 1:500) , or Biotinylated goat anti-rabbit antibody (Vector PK-6101 , 1:400) .	anti-TH		
To quantify the specificity of tdTomato expression in PVs and CHIs , we used antibodies against PV (rabbit anti-PV , SWANT PV25 , 1:1,000) or Chat (goat anti-Chat antibody , Abcam , No. ab254118 1:500) , followed by either Alexa Fluor 633 goat anti-rabbit secondary antibody (Invitrogen , No. A21070 , 1:1000	anti-PV , SWANT PV25		
	anti-Chat		
	anti-rabbit		
) for PV staining or Alexa Fluor 633 donkey anti-goat secondary antibody (Thermo Fisher , No. A21082 , 1:200	anti-goat		
Experimental Models: Organisms/Strains			
Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request EXPERIMENTAL MODEL AND SUBJECT DETAILS Adult GM24Gsat (Chat-cre, MMRRC Stock Number: 017269-UCD), B6 129P2-Pvalbtm1 (cre)ArbrJ (PV-cre, Jax Stock Number: 017320) and C57B6 mice (Charles River Laboratories, Cat C57BL6) were purchased from Charles River Laboratories,	GM24Gsat		
	017269-UCD), B6 129P2-Pvalbtm1 (cre)ArbrJ		
	C57B6		Suggestion: RRID:IMSR_JAX:000664(link)
B6 129P2-Pvalbtm1 (cre)ArbrJ (PV-cre , Jax Stock Number: 017320 , n = 6) transgenic mice were used to examine PVs and their interactions with SPNs .	B6 129P2-Pvalbtm1 (cre)ArbrJ		
These 13 transgenic mice , and another 3 C57BL6 mice were used to examine SPNs .	C57BL6		Suggestion: RRID:IMSR_JAX:000664(link)
Software and Algorithms			
All original code has been deposited at GitHub https:github.comHanLabBUDopamine-Depletion-CellReportsreleasestagv1.0	Zenodo		Suggestion: ZENODO (RRID:SCR_004129)(link)

and Zenodo https:doi.org10.5281zenodo.5786809 and is publicly available as of the date of publication.			
Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request EXPERIMENTAL MODEL AND SUBJECT DETAILS Adult GM24Gsat (Chat-cre, MMRRC Stock Number: 017269-UCD), B6 129P2-Pvalbtm1 (cre)ArbrJ (PV-cre, Jax Stock Number: 017320) and C57B6 mice (Charles River Laboratories, Cat C57BL6) were purchased from Charles River Laboratories,	Charles River Laboratories		Suggestion: Charles River Laboratories (RRID:SCR_003792)(link)
A custom multi-threaded Python script was used to send packaged < dx , dy > motion sensor data at 100 Hz to the image acquisition computer via a RS232 serial link .	Python		Suggestion: python-bx (RRID:SCR_024202)(link)
For the multi-taper calculation , we converted the mtspecgramc function and its dependency functions from the MATLAB chronux package (Chronux Home , n	MATLAB		Suggestion: MATLAB, RRID:SCR_001622(link)
Neuron pairs with correlation values within 95 of the shuffled distributions were deemed randomly correlated (random pairs).	Neuron		

Other Entities Detected

Your Sentences	Recognized Entity
Statistical Tests	
Calcium event Pearson correlation calculation Correlation coefficient between a neuron pair was estimated as the Pearson correlation of the two binarized calcium event vectors of the neuron pair.	Pearson correlation
To compare changes in Pearson correlation coefficient values between different time periods before and after 6-OHDA injury, we first computed the average correlation coefficient of all neuron pairs in a session, and then performed ANOVA and mixed effect model as detailed below.	ANOVA
To compare the correlation coefficients pre- and post- L-Dopa injection, we used a Wilcoxon signed rank test.	Wilcoxon signed rank test
To determine if a cell population responded to movement or rotation onset, we performed two-sided paired t test, comparing the Z-scores of the normalized fluorescence across all sessions of a given cell type.	t test
Additionally, we used Fishers exact test, and Wilcoxon signed rank test as specified in each Figure.	Fishers exact test
Unresolved RRIDs	
Striatal brain slices were stained using a Rabbit anti-TH antibody (Abcam , No. AB112 , 1:250) followed by either Alexa Fluor 633 goat anti-rabbit secondary antibody (Invitrogen , No. A21070 , 1:500) , or Biotinylated goat anti-rabbit antibody (Vector PK-6101 , 1:400) .	AB112

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